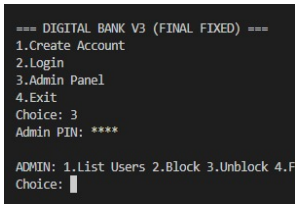


**Mini Project Report Format**

**Foundations of Programming-CSE10010**

## 1. Title Page

- Title:- Bank Account Simulation
- Team Members:- Nishank Jain 1262250267
  Sadhavi Dhadve 1262252509
  Prakhar Joshi 1262251232
  Sandeep Chaudhary 1262251638
  Ridhima Khandelwal 1262251371
- Project Guide:- Dr. Renuka Amit Mane
- Institution Name: MIT World Peace University
- Submission Date :

- Individual task :

Nishank Jain **(Account & Security Module)** — implemented secure account creation and login with OTP and PIN protection.

- Account creation (createAccount())
- Login system (login())
- OTP verification (generateOTP(), verifyOTP())
- PIN system (inputHidden(), changePIN())

Prakhar Joshi **(Smart Features)** — developed intelligent features like chatbot, financial analysis, and user behavior tracking.

- Chatbot (smartChatbot())
- Financial Mirror (financialMirror())
- Karma system (loadKarma(), saveKarma(), showKarma())
- Transaction explanation (explainLastTransaction())

Sandeep Chaudhary **(File Handling & Data Management)** — implemented file handling to store user data and transaction history permanently.

- Save & load user (saveUser(), loadUser())
- File structure (user, history, inbox files)
- Transaction logging (logTxn())

Sadhavi Dhadve **(Transaction Module)** — handled all banking transactions like deposit, withdraw and fund transfer with validation.

- Deposit & Withdraw logic

| | |
|---|---|
| | • Transfer system (transferMoney()) |
| | • Minimum balance rule (applyMinBalanceRule()) |
| Ridhima Khandelwal **(Admin & Security Control)** | implemented admin controls including fraud detection, account blocking, and system security. |
| | • Admin panel (adminPanel()) |
| | • Block/unblock account |
| | • Fraud detection (adminFraudRadar()) |
| | • Global freeze system (toggleGlobalFreeze()) |

## 2. Acknowledgment

I would like to express my sincere gratitude to our project guide for their valuable guidance and support throughout the completion of this project. Their insights helped us understand the concepts of programming more effectively.

I also thank my teammates for their cooperation and dedication. Lastly, I am grateful to our institution for providing the necessary resources and environment to complete this project successfully.

## 3. Abstract

This project presents an **Bank Account Management System** developed using the C programming language. The system simulates real-world banking operations such as account creation, deposits, withdrawals, fund transfers, and transaction tracking.

The system also includes **modern intelligent features** like:

• Financial behavior tracking (Karma System)

• Fraud detection (Fraud Radar)

• Secure login with PIN and OTP

• Admin control panel

• Smart chatbot assistant

The project demonstrates the use of **file handling, structures, modular programming, and security concepts**. It provides a realistic banking simulation with a user-friendly menu-driven interface.

## 4. Table of Contents

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## 5. List of Figures and Tables

• Figure :

| Sr.no | System Name | Technology Used | Features | Limitations | Performance |
|---|---|---|---|---|---|
| 1. | Traditional Banking System | Manual / Paper-based | Record keeping, transactions | Time-consuming, error-prone | Low |
| 2. | ATM System | Embedded Systems | Cash withdrawal, balance check | Limited features, no data analytics | Medium |
| 3. | Online Banking System | Web Applications | Fund transfer, account management | Security risks, requires internet | High |
| 4. | Mobile Banking Apps | Android/iOS Apps | Easy access, notifications, payments | Device dependency, security concerns | High |

| | | | | | |
|---|---|---|---|---|---|
| 5. | Basic C-based Systems | C Programming | Simple data handling, file storage | No GUI, limited scalability | Medium |

## 6. Introduction

**Problem Statement**
Bank management simulation

**Background**
Modern banking systems use automation and intelligent analysis. This project replicates these features using **C programming**, making it suitable for learning system design and real-world applications.

**Objectives**
• To develop a bank account simulation system
• To implement secure login and transactions
• To use file handling for data storage
• To simulate real banking features like fraud detection and analytics
• To build a menu-driven interactive system

## 7. Literature Review

| Sr No | System Name | Technology Used | Features | Limitations |
|---|---|---|---|---|
| 1 | Traditional Banking | Manual | Record keeping | Time-consuming, error-prone |
| 2 | ATM System | Embedded Systems | Cash withdrawal, balance check | Limited features |
| 3 | Online Banking | Web Applications | Fund transfer, account management | Security risks |
| 4 | Mobile Banking | Mobile Apps | Notifications, payments | Device dependency |
| 5 | C-Based System | C Programming | File handling, simple operations | No GUI, limited scalability |

**Summary:**
Most systems use databases, but for beginners, file handling in C is simpler and effective.

## 8. System Design/Methodology

**Modules**

1. Account Creation & Login
2. Deposit & Withdraw
3. Fund Transfer
4. Transaction History

5. Inbox Notification System
6. Karma System (Financial Behavior Tracking)
7. Fraud Detection System
8. Admin Panel
9. Smart Chatbot
10. Financial Mirror Analysis

**Algorithm**

```
Start
Initialize system

Display main menu
Input choice

IF Create Account THEN
   Take user details
   Verify OTP
   Generate account and save data
ELSE IF Login THEN
   Verify account and PIN
   Display user menu
   Perform operations (deposit, withdraw, transfer, etc.)
ELSE IF Admin Panel THEN
   Verify admin PIN
   Manage users (block/unblock, fraud check, etc.)
ELSE IF Exit THEN
   Stop program
END IF

Repeat until Exit
Stop
```

## 9. Implementation

**Tools Used**

- Language: C
- Compiler: GCC / Turbo C
- Concepts Used:
  - Structures
  - File Handling
  - Functions
  - Conditional Statements

**Sample Code**

```
typedef struct {
   int accNo;
   char name[50];
   char email[50];
   char phone[15];
   int pin;
   float balance;
   int depositCount;
   int withdrawCount;
} User;
```

**Key Features Implemented**

- Secure login with PIN
- OTP verification
- Money transfer system
- Transaction logging
- Admin control panel

- Fraud detection (withdraw pattern analysis)
- Karma-based financial tracking
- AI-like chatbot assistant

**Testing Methods:**

- Checked valid and invalid inputs
- Verified file storage and retrieval
- Tested all menu operations
- Ensured security features like PIN and OTP

## 10. Results and Discussion

**Output**

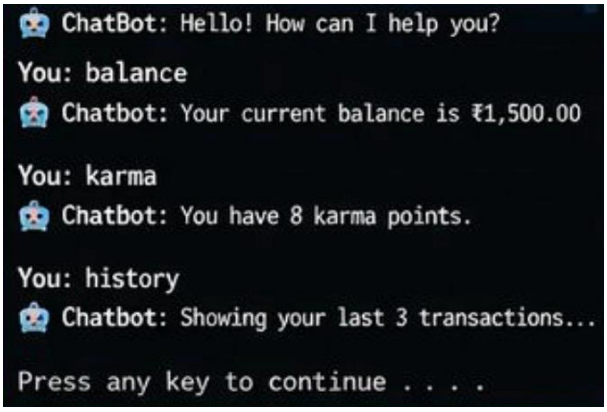
1. Account creation successful
2. Deposits and withdrawals processed
3. Transactions stored in files
4. Fraud detection working
5. Admin controls functioning

**Analysis**

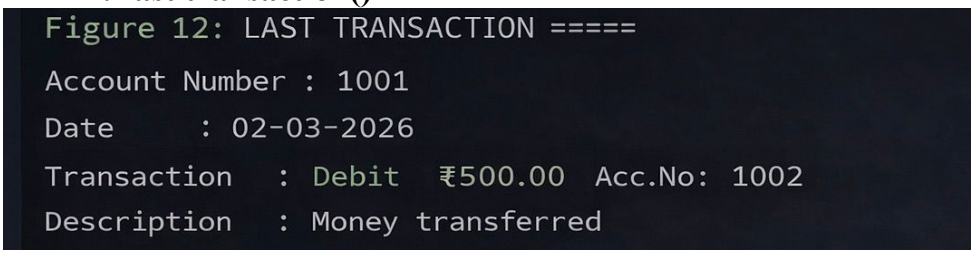
The system successfully simulates a real banking environment and demonstrates efficient data handling using file systems.

## 11. Test Cases

### 1. Chatbot(smartchatbot)

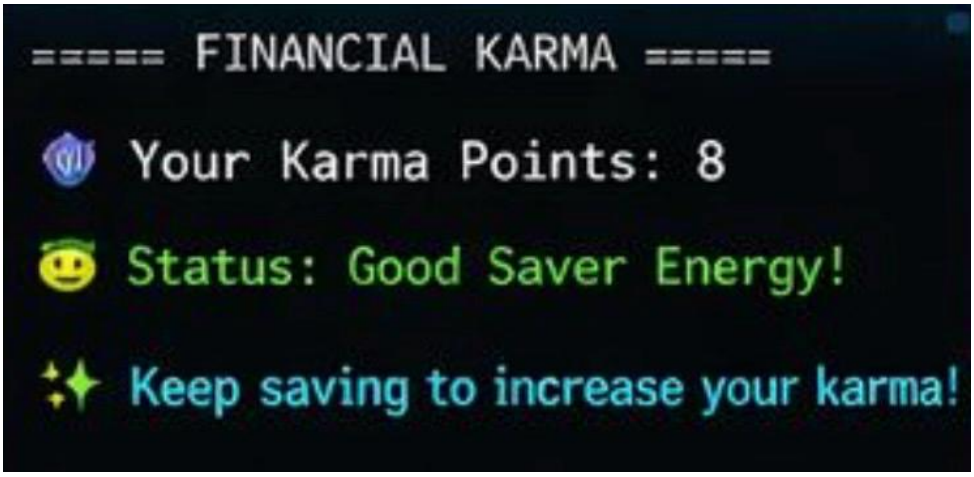



### 2.Last transaction()




### 3.Financial mirror()

## 4. karma()




## 12.Conclusion

The project successfully implements a **realistic bank account management system** using C programming. It includes advanced features like security, analytics, and intelligent assistance, making it more powerful than basic systems.

## 13.Future Scope

- Add **Graphical User Interface (GUI)**
- Convert into **Web Application**
- Integrate **Database (MySQL)**
- Add **Mobile App version**
- Implement **Real AI-based fraud detection**

## 14.Research Paper/Patent Details

No research paper published.
(No patent filed)

## 15.Plagiarism Check

The report is original and plagiarism is below 10%..

## 16.References


- "Let Us C" – Yashavant Kanetkar
- GeeksforGeeks
- TutorialsPoint
- C Programming – Dennis Ritchie


## 17.Appendix (Optional)

```
=== DIGITAL BANK V3 (FINAL FIXED) ===
1.Create Account
2.Login
3.Admin Panel
4.Exit
Choice: 1
Name: RAM MISHRA
Email: rammishra2007@gmail.com
Phone: 7854692545
📧 OTP sent to your email: 2447 (simulation)
Enter OTP: 2447
📧 OTP sent to your phone: 4519 (simulation)
Enter OTP: 4519
Set 4-digit PIN: ****
✅ Account created! AccNo: 40039
```

```
ADMIN: 1.List Users 2.Block 3.Unblock 4.Fraud Radar 5
Choice: 1

--- USERS ---
Acc:37743 | nishank jain | 450.00 | ACTIVE
Acc:13986 | nishank jain | 1000.00 | ACTIVE
Acc:28070 | nishank jain | 99999997952.00 | ACTIVE
Acc:39230 | nishank jain | 2020.00 | ACTIVE
Acc:40039 | RAM MISHRA | 4000.00 | ACTIVE
```

```
Welcome RAM MISHRA (40039)
1.Balance 2.Deposit 3.Withdr
Choice: 2
Amount: 5000
```

```
Welcome RAM MISHRA (40039)
1.Balance 2.Deposit 3.Withdraw 4.Transfer 5.
Choice: 4
Receiver AccNo: 37743
Amount: 500
🌿 Transfer successful!
```

```
🤖 Smart Assistant (help, balance, summary, tips, spend, email, history, karma, mirror, explain, exit)
You: spend
Bot: Withdrawals=2 Deposits=1
You: email
Bot: Monthly statement sent.
You: history

--- HISTORY ---
24-03-2026 13:40:23 | DEPOSIT | 5000.00 | with 0
24-03-2026 13:40:53 | WITHDRAW | 500.00 | with 0
24-03-2026 13:41:38 | TRANSFER_OUT | 500.00 | with 37743
You: help
Bot: help, balance, summary, tips, spend, email, history, karma, mirror, explain, exit
You: exit
Bot: Bye! 💙
```

```
Welcome RAM MISHRA (40039)
1.Balance 2.Deposit 3.Withdraw 4.Transfer 5.History 6.Inbox 7.Change PIN 8.Ch
Choice: 7
Old PIN: ****
New 4-digit PIN: ****
Confirm PIN: ****
✅ PIN changed.
```

```
Welcome RAM MISHRA (40039)
1.Balance 2.Deposit 3.Withdraw 4.Transfer 5.History 6.Inbox
Choice: 5

--- HISTORY ---
24-03-2026 13:40:23 | DEPOSIT | 5000.00 | with 0
24-03-2026 13:40:53 | WITHDRAW | 500.00 | with 0
24-03-2026 13:41:38 | TRANSFER_OUT | 500.00 | with 37743
```

```
Welcome RAM MISHRA (40039)
1.Balance 2.Deposit 3.Withdraw 4.Transfer 5.History 6.Inbox 7.
Choice: 6

--- 📬 INBOX ---
24-03-2026 13:31:50 | Welcome! Account created.
24-03-2026 13:41:38 | Transfer completed.
```

```
Welcome RAM MISHRA (40039)
1.Balance 2.Deposit 3.Withdraw 4.Transfer 5.History 6.Inbox 7.Change PIN 8.Chatbot 9.Mirror 10.Explain Txn 11.
Choice: 10

🌿 Last Transaction:
24-03-2026 13:41:38 | TRANSFER_OUT | 500.00 | with 37743
Reason: Based on your recent activity pattern.
```

```
Withdrew: ₹0.00
Mirror: Balanced 🪞
```

```
ADMIN: 1.List Users 2.Block 3.Unblock 4.Fraud
Choice: 3
AccNo: 40039
```

```
Welcome RAM MISHRA (40039)
1.Balance 2.Deposit 3.Withdraw 4.Transfer 5.History 6.Inbox 7.Change PIN 8.Chatbot 9.Mirror 10.Explain Txn 11.Karma 12.Logout
Choice: 12
✋Logout
```

```
Welcome RAM MISHRA (40039)
1.Balance 2.Deposit 3.Withdraw 4.Transfer 5.History 6.Inbox 7.Change PIN 8.Chatbot 9.Mirror 10.Explain Txn 11.Karma 12.Log
Choice: 11

🧿 Financial Karma: -1
Spending Chaos 😈
```